Random variables and distributions

CE/EL/MC/CV/GL/GM/ES 361 Probability and Statistics • First Semester, 2026/2027

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Week 3 • Chapter 3, Random Variables and Distributions

How to use this sheet

Try every question before the tutorial, in your group if you like, and bring your working on paper. Several of these questions have no single correct answer: they ask you to take a position and defend it with the numbers. A one line answer with no reasoning earns nothing, and a wrong answer argued carefully is worth more than a right answer copied from a friend.

Q1. For each quantity, say whether the random variable is discrete or continuous.

Number of trucks arriving at the crusher in an hour	Diameter of a drill core, in mm
Number of blows in a standard penetration test	Time until a pump fails
Ore grade of a sample, in g/t	Number of cubes failing out of 20 tested
Rainfall in a day, recorded to the nearest mm	Resistance of a resistor, measured to 3 decimals

(a) Two of these are really recorded as a short list of numbers but are almost always modelled as continuous. Which, and why is that acceptable?

(b) One of them is a count, yet an engineer might still model it with a continuous distribution. Which one, and when would that stop being safe?

Q2. A crate holds 10 pumps, of which 3 are faulty. Three pumps are drawn at random, without replacement, for a site job. Let X be the number of faulty pumps drawn.

(a) Derive the probability mass function of X and present it as a table.

(b) Check that it sums to 1, and find $P(X \geq 1)$.

(c) Explain why this is not a binomial situation, and describe the change to the experiment that would make it one.

(d) If the crate held 1 000 pumps with 300 faulty, would the difference still matter? Say roughly why.

Q3. A discrete random variable has $p(x) = kx$ for $x = 1, 2, 3, 4$ and zero elsewhere.

(a) Find k.

(b) Find $P(X \geq 3)$ and $P(X$ is even$)$.

(c) Write down the cumulative distribution function and sketch it.

(d) Why must the sketch be a staircase and not a smooth curve? What does the height of each step represent?

Q4. The number of emergency stops X on a conveyor in a shift has the following cumulative distribution function.

x	0	1	2	3	4
$F(x)$	0.10	0.35	0.60	0.90	1.00

(a) Recover the probability mass function.

(b) Find $P(1 < X \leq 3)$ and $P(X \geq 2)$.

(c) A supervisor reads $F(2) = 0.60$ as "there is a 60% chance of exactly two stops". Correct him.

(d) What is $P(X = 2.5)$, and what does your answer say about the kind of variable this is?

Q5. The depth of wear X, in mm, on a liner plate after one month has density

$$f(x) = c\,x(2-x), \qquad 0 \le x \le 2,$$

and zero elsewhere.

(a) Find c.

(b) Sketch the density. What does its shape say about wear on this plate?

(c) Find $P(X > 1.5)$ and $P(0.5 < X < 1.5)$.

(d) Find the median wear. Do it without integrating, then confirm by integration.

(e) The plate is replaced when wear passes 1.5 mm. In a yard of 200 plates, roughly how many need replacing after one month, and what assumption are you making about the plates?

Q6. A student writes: "$f(x) = x/2$ for $0 \le x \le 3$ is a probability density function, and since $f(1.2) = 0.6$, the probability that $X = 1.2$ is 0.6."

(a) Show that f is not a density, and find the constant that would fix it.

(b) The second half of the sentence is wrong twice over. Explain both errors.

(c) Once corrected, find $P(X > 2)$.

(d) Can a density ever take a value greater than 1? Give an example or explain why not.

Q7. Trucks arrive at a weighbridge at random, on average one every 12 minutes, and the waiting time T between arrivals follows $f(t) = \lambda e^{-\lambda t}$ for $t \ge 0$.

(a) State λ and find $P(T > 20)$.

(b) You have already waited 10 minutes. Find the probability you wait at least 20 minutes more.

(c) Compare your two answers. State the property this illustrates, and say whether you find it believable for trucks.

(d) The weighbridge operator says "no truck for 25 minutes, so one is well overdue". Is he right? Where would his reasoning be right instead?

Q8. Daily rainfall is recorded to the nearest millimetre and modelled as a continuous random variable.

(a) Under the model, $P(X = 5) = 0$, yet the rain gauge reads exactly 5 mm on many days. Resolve the contradiction.

(b) Express "the gauge reads 5 mm" as an event about the underlying continuous variable.

(c) Explain why, for a continuous variable, $P(X \ge a)$ and $P(X > a)$ are the same, while for the conveyor stops in Q4 they are not.

Q9. A handheld GPS reports eastings with an error that is modelled as uniform on $[-3, 3]$ metres.

(a) Sketch the density and find $P(|E| > 2)$ and $P(|E| < 0.5)$.

(b) Under this model, an error of 2.99 m is exactly as likely as an error of 0.01 m. Is that plausible for a real instrument?

(c) Sketch the density shape you would expect instead, and name the feature of the uniform model that is wrong.

(d) Give one situation in surveying where a uniform model really is the right choice.

Q10. For each quantity below, sketch the density you would expect and write one sentence defending the shape: where the peak is, whether it is symmetric, and whether it has a long tail.

(a) Time to failure of a pump that is most likely to fail early in its life.

(b) Error in a well calibrated instrument.

(c) Ore grade in a deposit made of two different rock types.

(d) Time of day at which a mill trips.

Then say which of the four you would be uncomfortable describing by a mean and a standard deviation alone, and why.

A sketch is an answer here, not decoration. Label the axes, mark where the mass sits, and make sure the area under any density you draw could plausibly be 1.